

LUCAS TENDICK



- Postdoctoral researcher at Inria Paris-Saclay since 01/12/2023
- Currently working on quantum resources and correlations in networks
- Interested in the fundamental differences between classical and quantum physics
- MSCA Postdoctoral Fellow starting from 03/2026

CONTACT

- ✉ lucas-amadeus.tendick@inria.fr
- 📍 91120 Palaiseau, France
- 👤 Nationality: German
- 📅 Date of birth: 05/11/1994
- 🆔 0000-0001-5210-6710 (Orcid ID)
- 🎓 Scholar
- 📄 ArXiv submission list
- 🐙 @LucasTendick (GitHub)

RESEARCH INTERESTS

- Quantum correlations
- Entanglement, steering, nonlocality
- Quantum networks
- Quantum information processing
- Quantum resource theories
- Quantum incompatibility
- Quantum measurement theory
- Simulability of quantum devices
- Quantum foundations
- Tests of quantum mechanics
- General causal structures

SKILLS

German (mother tongue)	●●●●●●
English	●●●●●●
French	●●●●●●
Matlab	●●●●●●
Semidefinite programming	●●●●●●
CVX	●●●●●●
YALMIP	●●●●●●
Mathematica	●●●●●●
Python	●●●●●●
LaTeX	●●●●●●
Office	●●●●●●
Windows	●●●●●●
Linux	●●●●●●
Responsible	●●●●●●
Organized	●●●●●●
Independence	●●●●●●
Student supervision	●●●●●●
Teaching assistance	●●●●●●

CURRENT POSITION

- 📅 01/12/2023-Now
- 📍 Inria Paris-Saclay, Bâtiment Alan Turing, 1 rue Honoré d'Estienne d'Orves 91120 Palaiseau
PhiQus Group
Marc-Olivier Renou
- Postdoctoral researcher

EDUCATION

- 📅 02/01/2019-23/03/2023
- 📍 Institute for Theoretical Physics III, Heinrich Heine University Düsseldorf, Dagmar Bruß group
- PhD in Physics

PhD Thesis:

- Title: Interplay of Quantum Resources in Bell-type Scenarios
- Overall grade: Magna cum laude
- Supervisor: Dagmar Bruß
- Co-supervisor: Hermann Kampermann

- 📅 02/2017- 12/2018
- 📍 Heinrich Heine University Düsseldorf
- Supervisor: Dagmar Bruß
- Thesis title: Hidden nonlocality
- Overall grade: 1.1 (Grading scale: 1.0 best, 4.0 worst)
- MSc in Physics

- 📅 10/2013 - 01/2017
- 📍 Heinrich Heine University Düsseldorf
- Supervisor: Dagmar Bruß
- Thesis title: Quantum clocks
- Overall grade: 1.8 (Grading scale: 1.0 best, 4.0 worst)
- BSc in Physics

- 📅 08/2005 - 07/2013
- 📍 Gymnasium Gerresheim, Düsseldorf
- Overall grade: 1.6 (Grading scale: 1.0 best, 4.0 worst)
- School organizations: Chess club (participation at tournaments)
- High-school Diploma

TEACHING ASSISTANCE

- Mathematical methods for physics (Düsseldorf)
- Theoretical classical mechanics (Düsseldorf)
- Quantum mechanics I (Düsseldorf)
- Physics for pharmacists (Düsseldorf)
- Organization Group Seminars (Düsseldorf and Paris-Saclay)
- Advanced quantum information (École polytechnique)

- Bachelor
 - B. Becker, *Optimal local filters for two-qubit systems regarding Bell inequalities* (2019)
 - H. Kapetanovic, *Quantum speed limits* (2020)
 - P. Braun, *Hierarchy of quantum resources* (2023)
 - M. Zehnpfennig, *On the power of POVMs versus projective measurements* (2023)
 - O. Cohausz, *Informativeness of an assemblage* (2023)
- Master
 - C. Helle, *Purity relations in composite systems* (2022)
 - T. Glorieux, *Coloring problems in distributed computing* (2024, Internship)
 - A. Coquet *Shared Random bits in causal structures* (2024, Internship)
- PhD
 - X. Xu (ongoing)
 - F. Moradi (ongoing)
 - A. Coquet (ongoing)

CONFERENCES, WORKSHOPS, SERVICES, AND OTHERS

- Conferences, Workshops, and Seminars
 - 851. WE-Heraeus-Seminar, Bad Honnef, April 2026, **Poster**
 - Distributed Quantum Computing Workshop, October 2025, Helsinki, **invited talk**
 - YQIS 2025, Barcelona, October 2025, **contributed talk**
 - CEQIP 2025, Smolenice, July 2025, **Poster**
 - 828. WE-Heraeus-Seminar, Bad Honnef, March 2025, **Poster**
 - DPG Spring Meeting, Bonn, March 2025
 - GdR TeQ, Paris, November 2024, **contributed talk**
 - Vienna Quantum Foundations Conference, Vienna September 2024
 - Quantum Information Working Group Seminar Warsaw, April 2024, **invited online talk**
 - Doubts and Beliefs in Quantum Information, Düsseldorf, April 2024
 - DPG Spring Meeting, Berlin, March 2024, **contributed talk**
 - QUACC, Warsaw, November 2023, **contributed talk**
 - Quantum Corona 2 Workshop, Siegen, Oktober 2023, **contributed talk**
 - DPG Spring Meeting, Hannover, March 2023, **contributed talk**
 - Quantum Corona Workshop, Siegen, September 2022, **contributed talk**
 - DPG Spring Meeting, Regensburg, September 2022, **contributed talk**
 - 741. WE-Heraeus-Seminar, Bad Honnef, Juli 2022, **Poster**
 - DPG Spring Meeting (virtual), September 2021, **Poster**
 - Symposium on Mathematical Physics, Torun (virtual), June 2021
 - QIP (virtual), February 2021
 - Q-Turn (virtual), November 2020
 - DPG Fall Meeting, Freiburg, September 2019, **Poster**
 - The Mathematics of Quantum Information Workshop, Siegen, March 2019
 - DPG Spring Meeting, Rostock, March 2019, **Poster**
- Group visits and (online) talks
 - Nordita Stockholm, Roope Uola, Oktober 2025
 - ICFO, Quantum Information Theory Group, Antonio Acín, Oktober 2025
 - Theoretical Quantum Optics Group Siegen, Matthias Kleinmann, May 2025
 - Lund University, Armin Tavakoli, January 2025
 - Institute for Quantum Inspired and Quantum Optimization in Hamburg, Martin Kliesch, January 2025
 - Quantum information theory group Düsseldorf, Dagmar Bruß, December 2024
 - Quantum information theory group Cologne, David Gross, March 2024
 - Theoretical Quantum Optics Group Siegen, Otfried Gühne, March 2024
 - Quantum information theory group Düsseldorf, Dagmar Bruß, March 2024

- LIP6 - QI Team in Paris, Marco Túlio Quintino, September 2023
- INRIA Saclay, Marc-Olivier Renou, September 2023
- Quantum information theory group in Bristol, Paul Skrzypczyk, March 2023
- Quantum theory group in Glasgow (online), Sarah Croke, March 2023
- Service
 - Referee for PRL
 - Referee for PRA
 - Referee for PR Applied
 - Referee for PRR
 - Referee for PRX Quantum
 - Referee for Quantum
 - Referee for NPJ Quantum Information
- Grants
 - Granted MSCA fellowship in the September 2024 call
 - Application for a Newton International Fellowship grant with Sarah Croke, 2023, (no fellowship was granted)
- Organization
 - Organizer of the YQIS24 conference
 - Steering Committee Member of YQIS, October 2025-now
- Others
 - Final-four candidate of the Wilhelm und Else Heraeus Dissertationprice 2023 between HHU Düsseldorf and BU Wuppertal

PUBLICATIONS

1. Activation of nonlocality in bound entanglement

 **L. Tendick**, H. Kampermann, and Dagmar Bruß

 2020  Phys. Rev. Lett. 124, 050401

 [APS](#), [arXiv](#)

2. Quantifying necessary quantum resources for nonlocality

 **L. Tendick**, H. Kampermann, and Dagmar Bruß

 2022  Phys. Rev. Research 4, L012002

 [APS](#), [arXiv](#)

3. Distance-based resource quantification for sets of quantum measurements

 **L. Tendick**, M.Kliesch, H. Kampermann, and Dagmar Bruß

 2023  Quantum 7, 1003 (2023)

 [Quantum](#), [arXiv](#)

4. Distributed quantum incompatibility

 **L. Tendick**, H. Kampermann, and Dagmar Bruß

 2023  Phys. Rev. Lett. 131, 120202

 [APS](#), [arXiv](#)

5. Quantum Correlations Cannot Be Reproduced with a Finite Number of Measurements in Any No-Signaling Theory

 **L. Tendick**

 2024  Phys. Rev. Lett. 135, 030201

 [APS](#), [arXiv](#)

6. Distributed Quantum Advantage for Local Problems

 A. Balliu, S. Brandt, X. Coiteux-Roy, F. d'Amore, M. Equi, F. Le Gall, H. Lievonen, A. Modanese, D. Olivetti, M. Renou, J. Suomela, **L. Tendick**, I. Veeren,
 2025  Stoc ' 25, 451-462  [STOC, arXiv](#)

7. Unlimited quantum correlation advantage from bound entanglement

 A. Tavakoli, C. R. i Carceller, **L. Tendick**, T. Vértesi
 2025  [arXiv:2504.06395](#)  [arXiv](#)

8. Strict hierarchy between n -wise measurement simulability, compatibility structures, and multi-copy compatibility

 **L. Tendick**, C. Budroni, M. T. Quintino
 2025  [arXiv:2506.21223](#)  [arXiv](#)

9. Quantitative Quantum Soundness for Bipartite Compiled Bell Games via the Sequential NPA Hierarchy

 I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, **L. Tendick**, X. Xu, Y. Zhao
 2025  [arXiv:2507.17006](#)  [arXiv](#)

10. Quantitative quantum soundness for all multipartite compiled nonlocal games

 M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, **L. Tendick**, X. Xu
 2025  [arXiv:arXiv:2509.25145](#)  [arXiv](#)

11. Can every set of incompatible measurements lead to genuine multipartite steering?

 L. EA Porto, **L. Tendick**, D. Cavalcanti, R. Uola, M.T. Quintino
 2026  [arXiv:2603.25345](#)  [arXiv](#)

12. Coordination Requires a Common Cause in Quantum Theory

 D. Centeno, A. Coquet, M.C. Alañón, **L. Tendick**, M. Renou, E. Wolfe
 2026  [arXiv:2605.03120](#)  [arXiv](#)

13. A missing causal principle: Coordination

 D. Centeno, A. Coquet, M.C. Alañón, **L. Tendick**, M. Renou, E. Wolfe
 2026  [arXiv:2605.03132](#)  [arXiv](#)

REFEREES

1. Prof. Dr. Dagmar Bruß
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3. Prof. Dr. Martin Kliesch
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Hamburg University of Technology
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2. Jun. Prof. Dr. Marc-Olivier Renou
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